

CUSTOMER SEGMENTATION USING K-MEANS CLUSTERING

Internship Project Report

Project Title: Customer Segmentation Using Machine Learning

Organization: Thiranex

Technology: Python, Jupyter Notebook, Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn

Machine Learning Algorithm: K-Means Clustering

Dataset Size: 1,000 Customer Records

1. ABSTRACT

Customer segmentation is an important business analytics technique used to divide customers into groups with similar characteristics, purchasing behavior, and engagement patterns.

The objective of this project is to segment customers using machine learning techniques. A customer dataset containing 1,000 records was analyzed using Python. The dataset contains demographic information, income, spending behavior, purchasing frequency, purchase value, online purchasing behavior, product preferences, membership level, discount usage, recency, and customer satisfaction.

The project follows a complete data science workflow consisting of data collection, data understanding, data cleaning, exploratory data analysis, feature selection, feature scaling, clustering, model evaluation, visualization, and business interpretation.

The K-Means clustering algorithm was selected to create customer segments. The Elbow Method and Silhouette Score were used to determine a suitable number of clusters. Principal Component Analysis (PCA) was then used to visualize the resulting customer segments.

The resulting customer groups can be used by businesses to develop targeted marketing campaigns, improve customer retention, personalize offers, identify high-value customers, and increase customer lifetime value.

2. INTRODUCTION

Modern businesses interact with customers having different needs, purchasing habits, income levels, preferences, and levels of engagement.

Treating every customer in the same way may result in inefficient marketing and poor customer engagement. Customer segmentation solves this problem by grouping customers according to similarities in their characteristics and behavior.

Machine learning provides an effective approach to automatically discover these groups.

In this project, K-Means clustering is used to divide customers into meaningful groups based on numerical customer attributes.

3. PROBLEM STATEMENT

Businesses have a large number of customers with different demographic and purchasing characteristics.

The main problem is:

How can customers be automatically grouped into meaningful segments based on their demographic characteristics, spending behavior, purchasing frequency, engagement, and satisfaction?

The identified segments can then be used to design targeted business and marketing strategies.

4. PROJECT OBJECTIVES

The major objectives of this project are:

1. To understand customer demographic and behavioral data.
2. To clean and preprocess the customer dataset.
3. To perform exploratory data analysis.
4. To identify important customer characteristics.
5. To standardize numerical features.
6. To determine a suitable number of customer clusters.
7. To apply K-Means clustering.
8. To evaluate the clustering results.
9. To visualize customer segments using PCA.
10. To profile each customer segment.
11. To provide business recommendations based on the segments.
12. To create a final dataset containing customer segment labels.

5. DATASET DESCRIPTION

The dataset contains 1,000 customer records.

The main attributes are:

Feature	Description
CustomerID	Unique customer identifier
Age	Customer age
Gender	Customer gender
AnnualIncome	Annual customer income
SpendingScore	Spending score between 1 and 100
PurchaseFrequency	Number of purchases
AvgPurchaseAmount	Average purchase amount
TotalPurchaseValue	Total purchase value
OnlinePurchasePercentage	Percentage of online purchases
RecencyDays	Number of days since recent purchase
PreferredCategory	Preferred product category
Location	Customer location
MembershipLevel	Customer membership level
DiscountUsagePercentage	Percentage of purchases using discounts
SatisfactionScore	Customer satisfaction score

6. TOOLS AND TECHNOLOGIES

The following technologies were used:

Python

Used as the primary programming language for data analysis and machine learning.

Jupyter Notebook

Used as the interactive development environment.

Pandas

Used for loading, cleaning, manipulating, and analyzing the dataset.

NumPy

Used for numerical operations.

Matplotlib

Used for visualization.

Seaborn

Used for statistical visualization.

Scikit-learn

Used for:

- StandardScaler
- KMeans
- Silhouette Score
- PCA

7. PROJECT WORKFLOW

The project follows the following workflow:

Dataset

↓

Data Understanding

↓

Data Cleaning

↓

Exploratory Data Analysis

↓

Feature Selection

↓

Feature Scaling

↓

Elbow Method

↓

Silhouette Score

↓

K-Means Clustering

↓

PCA Visualization



Cluster Profiling



Business Recommendations

8. DATA LOADING

The dataset was loaded using Pandas.

import pandas as pd

```
df = pd.read_csv("customer_segmentation_1000.csv")
```

```
print(df.shape)
```

```
df.head()
```

The dataset contains:

- **1,000 rows**
- **15 original columns**

9. DATA UNDERSTANDING

The dataset was examined using:

```
df.info()
```

and:

```
df.describe()
```

The analysis included:

- Data types
- Number of observations
- Numerical statistics
- Categorical variables
- Missing values

- Duplicate records

10. DATA CLEANING

The following preprocessing operations were performed.

10.1 Missing Value Check

```
df.isnull().sum()
```

Missing numerical values, if present, were handled using median values.

Categorical missing values were handled using the mode.

10.2 Duplicate Check

```
df.duplicated().sum()
```

Duplicate records were removed using:

```
df = df.drop_duplicates()
```

This prevents duplicate customers from influencing the clustering algorithm.

11. EXPLORATORY DATA ANALYSIS

Exploratory Data Analysis was performed to understand customer characteristics and behavior.

The main analyses included:

- Age distribution
- Income distribution
- Spending score distribution
- Gender distribution
- Product category distribution
- Membership distribution
- Correlation analysis

12. CUSTOMER AGE DISTRIBUTION

The age distribution helps understand the demographic structure of the customer base.

```
plt.figure(figsize=(8,5))
```

```
sns.histplot(df["Age"], bins=15, kde=True)
plt.title("Customer Age Distribution")
plt.xlabel("Age")
plt.ylabel("Number of Customers")
plt.show()
```

Figure 1: Customer Age Distribution

The graph shows the distribution of customers across different age groups.

Age is included as one of the clustering variables because customer purchasing behavior can differ across age groups.

13. ANNUAL INCOME DISTRIBUTION

Annual income is an important variable because purchasing capacity can be related to income.

```
plt.figure(figsize=(8,5))
sns.histplot(df["AnnualIncome"], bins=20, kde=True)
plt.title("Annual Income Distribution")
plt.xlabel("Annual Income")
plt.ylabel("Number of Customers")
plt.show()
```

Figure 2: Annual Income Distribution

This graph provides an overview of the income levels represented in the dataset.

14. SPENDING SCORE DISTRIBUTION

Spending Score is one of the most important behavioral variables.

```
plt.figure(figsize=(8,5))
sns.histplot(df["SpendingScore"], bins=20, kde=True)
plt.title("Spending Score Distribution")
plt.xlabel("Spending Score")
plt.ylabel("Number of Customers")
```

```
plt.show()
```

Figure 3: Spending Score Distribution

Customers have different levels of spending behavior, making this feature useful for segmentation.

15. PREFERRED PRODUCT CATEGORY

Customers have different product preferences.

```
plt.figure(figsize=(10,5))  
sns.countplot(  
    data=df,  
    x="PreferredCategory",  
    order=df["PreferredCategory"].value_counts().index  
)  
plt.title("Preferred Product Category")  
plt.xlabel("Product Category")  
plt.ylabel("Number of Customers")  
plt.xticks(rotation=30)  
plt.show()
```

Figure 4: Preferred Product Category

This analysis helps businesses understand which product categories are preferred by customers.

16. CORRELATION ANALYSIS

A correlation heatmap was created to identify relationships between numerical variables.

```
numeric_columns = df.select_dtypes(include=np.number).columns  
plt.figure(figsize=(12,8))  
sns.heatmap(  
    df[numeric_columns].corr(),  
    annot=True,  
    cmap="coolwarm",
```



```
fmt=".2f"  
)  
plt.title("Correlation Heatmap")  
plt.show()
```

Figure 5: Correlation Heatmap

The heatmap helps identify relationships between variables such as:

- Annual Income
- Spending Score
- Purchase Frequency
- Purchase Value
- Recency
- Satisfaction

These relationships are useful when interpreting the final customer segments.

17. FEATURE SELECTION

The following numerical variables were selected for clustering:

```
features = [  
    "Age",  
    "AnnualIncome",  
    "SpendingScore",  
    "PurchaseFrequency",  
    "AvgPurchaseAmount",  
    "TotalPurchaseValue",  
    "OnlinePurchasePercentage",  
    "RecencyDays",  
    "DiscountUsagePercentage",  
    "SatisfactionScore"  
]
```

These variables represent both demographic and behavioral characteristics.

18. FEATURE SCALING

K-Means is a distance-based algorithm.

The selected variables have different scales.

For example:

- Age is measured in years.
- Income may be measured in thousands.
- Spending Score ranges from 1 to 100.
- Purchase Value may be in thousands.

Therefore, StandardScaler was used.

```
from sklearn.preprocessing import StandardScaler
```

```
scaler = StandardScaler()
```

```
X_scaled = scaler.fit_transform(df[features])
```

Scaling prevents variables with large numerical values from dominating the clustering process.

19. K-MEANS CLUSTERING

K-Means is an unsupervised machine learning algorithm that divides observations into K groups.

The algorithm works by:

1. Selecting K initial cluster centers.
2. Assigning each observation to the nearest center.
3. Recalculating cluster centers.
4. Repeating the process until the clusters stabilize.

The objective is to minimize the within-cluster sum of squared distances.

20. ELBOW METHOD

The Elbow Method was used to examine different values of K.

```
from sklearn.cluster import KMeans
```

```
inertia = []
```

```
for k in range(2, 11):
```

```

kmeans = KMeans(
    n_clusters=k,
    random_state=42,
    n_init=10
)
kmeans.fit(X_scaled)
inertia.append(kmeans.inertia_)

```

The resulting values were plotted:

```

plt.figure(figsize=(9,5))
plt.plot(
    range(2,11),
    inertia,
    marker="o"
)
plt.title("Elbow Method for Optimal K")
plt.xlabel("Number of Clusters")
plt.ylabel("Inertia")
plt.xticks(range(2,11))
plt.grid(True)
plt.show()

```

Figure 6: Elbow Method

The Elbow Method helps identify the point where adding additional clusters produces diminishing improvements.

For this project, **K = 4** is used as the final segmentation solution because it provides a practical and interpretable four-segment structure.

21. SILHOUETTE SCORE

The Silhouette Score was also calculated to evaluate cluster separation.

```

from sklearn.metrics import silhouette_score

silhouette_scores = []

```

```

for k in range(2, 11):
    kmeans = KMeans(
        n_clusters=k,
        random_state=42,
        n_init=10
    )
    labels = kmeans.fit_predict(X_scaled)
    score = silhouette_score(
        X_scaled,
        labels
    )
    silhouette_scores.append(score)

```

The scores were visualized:

```

plt.figure(figsize=(9,5))
plt.plot(
    range(2,11),
    silhouette_scores,
    marker="o"
)
plt.title("Silhouette Score")
plt.xlabel("Number of Clusters")
plt.ylabel("Silhouette Score")
plt.xticks(range(2,11))
plt.grid(True)
plt.show()

```

Figure 7: Silhouette Score

A higher silhouette score indicates that observations are generally closer to their own cluster than to neighboring clusters.

The exact final score should be copied from the output of the notebook.

22. FINAL K-MEANS MODEL

The final model was created using four clusters.

```
kmeans = KMeans(  
    n_clusters=4,  
    random_state=42,  
    n_init=10  
)  
df["Cluster"] = kmeans.fit_predict(X_scaled)
```

Each customer now has an assigned cluster.

23. CUSTOMER COUNT BY CLUSTER

The number of customers in each cluster can be calculated using:

```
df["Cluster"].value_counts().sort_index()
```

Visualization:

```
cluster_counts = df["Cluster"].value_counts().sort_index()  
plt.figure(figsize=(8,5))  
sns.barplot(  
    x=cluster_counts.index,  
    y=cluster_counts.values  
)  
plt.title("Number of Customers in Each Cluster")  
plt.xlabel("Cluster")  
plt.ylabel("Number of Customers")  
plt.show()
```

Figure 8: Customer Count by Cluster

This graph shows how the 1,000 customers are distributed among the four segments.

Insert the exact cluster counts generated by your notebook here.

24. PCA VISUALIZATION

Because ten features were used for clustering, visualizing the clusters directly is difficult.

Principal Component Analysis (PCA) was therefore used to reduce the feature space to two dimensions.

```
from sklearn.decomposition import PCA
```

```
pca = PCA(n_components=2)
```

```
X_pca = pca.fit_transform(X_scaled)
```

```
df["PCA1"] = X_pca[:,0]
```

```
df["PCA2"] = X_pca[:,1]
```

The clusters were visualized:

```
plt.figure(figsize=(10,7))
```

```
sns.scatterplot(
```

```
    data=df,
```

```
    x="PCA1",
```

```
    y="PCA2",
```

```
    hue="Cluster",
```

```
    palette="viridis",
```

```
    s=70
```

```
)
```

```
plt.title("Customer Segmentation using K-Means")
```

```
plt.xlabel("Principal Component 1")
```

```
plt.ylabel("Principal Component 2")
```

```
plt.legend(title="Cluster")
```

```
plt.show()
```

Figure 9: PCA Customer Segmentation Plot

The PCA visualization provides a two-dimensional representation of the customer segments.

Customers belonging to the same cluster are expected to have similar characteristics across the selected clustering variables.

25. CLUSTER PROFILE

The most important step after clustering is understanding what each cluster represents.

```
cluster_profile = df.groupby("Cluster")[features].mean()
```

26. CUSTOMER SEGMENT INTERPRETATION

After examining the cluster profile, the four clusters can be given meaningful business names.

Segment 1 – High-Value Customers

These customers should be identified using characteristics such as:

- High income
- High spending score
- High purchase frequency
- High total purchase value
- Good satisfaction
- Strong engagement

Recommended strategy

- Premium loyalty programs
- Exclusive offers
- Early access to products
- Personalized recommendations
- VIP membership benefits
- Cross-selling and upselling

The main objective is to retain these customers and increase their lifetime value.

Segment 2 – Regular Customers

These customers show moderate purchasing behavior.

Recommended strategy

- Product bundles
- Reward points
- Cross-selling
- Personalized product recommendations

- Regular promotional campaigns

The objective is to gradually increase purchase frequency and average purchase value.

Segment 3 – Potential / Affluent Customers

These customers may have relatively strong income or purchasing capacity but may not yet demonstrate the highest engagement.

Recommended strategy

- Premium product recommendations
- Membership upgrades
- Personalized campaigns
- High-value product promotions
- Targeted loyalty incentives

The goal is to convert their potential into higher customer value.

Segment 4 – Low-Engagement Customers

These customers may show:

- Lower spending
- Lower purchase frequency
- Higher recency
- Lower overall purchase value

Recommended strategy

- Re-engagement campaigns
- Personalized discounts
- Limited-time offers
- Reminder campaigns
- Customer feedback surveys
- Special retention offers

The goal is to reactivate customers before they become permanently inactive.

27. INCOME BY CLUSTER

```
plt.figure(figsize=(9,5))
sns.boxplot(
    data=df,
    x="Cluster",
    y="AnnualIncome"
)
plt.title("Annual Income by Customer Cluster")
plt.xlabel("Cluster")
plt.ylabel("Annual Income")
plt.show()
```

Figure 10: Annual Income by Cluster

This visualization allows comparison of income levels across customer segments.

28. SPENDING SCORE BY CLUSTER

```
plt.figure(figsize=(9,5))
sns.boxplot(
    data=df,
    x="Cluster",
    y="SpendingScore"
)
plt.title("Spending Score by Customer Cluster")
plt.xlabel("Cluster")
plt.ylabel("Spending Score")
plt.show()
```

Figure 11: Spending Score by Cluster

This graph helps identify which segments have the strongest and weakest spending behavior.

29. PURCHASE FREQUENCY BY CLUSTER

```
plt.figure(figsize=(9,5))
sns.boxplot(
    data=df,
    x="Cluster",
    y="PurchaseFrequency"
)
plt.title("Purchase Frequency by Customer Cluster")
plt.xlabel("Cluster")
plt.ylabel("Purchase Frequency")
plt.show()
```

Figure 12: Purchase Frequency by Cluster

This graph provides insight into customer activity levels.

30. PRODUCT CATEGORY BY CLUSTER

The relationship between product category and customer segment was analyzed using:

```
category_cluster = pd.crosstab(
    df["Cluster"],
    df["PreferredCategory"]
)
```

category_cluster

Visualization:

```
category_cluster.plot(
    kind="bar",
    figsize=(12,6)
)
plt.title("Preferred Product Category by Cluster")
plt.xlabel("Cluster")
plt.ylabel("Number of Customers")
```

```
plt.xticks(rotation=0)
plt.show()
```

Figure 13: Preferred Product Category by Cluster

This analysis can help businesses customize product recommendations for each segment.

31. MEMBERSHIP LEVEL BY CLUSTER

Membership distribution was analyzed using:

```
membership_cluster = pd.crosstab(
    df["Cluster"],
    df["MembershipLevel"]
)
```

membership_cluster

Visualization:

```
membership_cluster.plot(
    kind="bar",
    stacked=True,
    figsize=(10,6)
)
plt.title("Membership Level Distribution by Cluster")
plt.xlabel("Cluster")
plt.ylabel("Number of Customers")
plt.xticks(rotation=0)
plt.show()
```

Figure 14: Membership Level by Cluster

This helps identify which customer segments contain more premium members.

32. FINAL CUSTOMER SEGMENT TABLE

The final customer dataset contains the original customer information along with the cluster assignment.

```
df[
    [
        "CustomerID",
        "Age",
        "AnnualIncome",
        "SpendingScore",
        "PurchaseFrequency",
        "TotalPurchaseValue",
        "Cluster"
    ]
].head(20)
```

A new business-friendly segment column can also be created:

```
cluster_names = {
    0: "High-Value Customers",
    1: "Regular Customers",
    2: "Potential / Affluent Customers",
    3: "Low-Engagement Customers"
}
```

```
df["CustomerSegment"] = df["Cluster"].map(cluster_names)
```

33. BUSINESS INSIGHTS

The customer segmentation analysis provides several useful business insights.

Insight 1: Customers are not homogeneous

Customers differ significantly in:

- Income
- Spending
- Purchase frequency
- Recency

- Product preferences
- Membership level
- Satisfaction

Therefore, a single marketing strategy may not be appropriate for all customers.

Insight 2: High-value customers should receive priority

Customers with high spending and high purchase value represent an important revenue opportunity.

Businesses should focus on retention and loyalty strategies for this group.

Insight 3: Low-engagement customers require reactivation

Customers showing low purchasing activity or long recency periods can be targeted using personalized campaigns.

Insight 4: Customer preferences can support personalization

Preferred product categories can be used to recommend products and create category-specific promotions.

Insight 5: Membership can be used for customer development

Regular and potential customers can be encouraged to move toward higher membership levels through rewards and loyalty programs.

34. MARKETING RECOMMENDATIONS

High-Value Customers

Recommended actions:

- VIP rewards
- Premium offers
- Exclusive products
- Early access
- Personalized recommendations
- Loyalty benefits

Regular Customers

Recommended actions:

- Bundled offers

- Reward points
- Cross-selling
- Seasonal campaigns
- Product recommendations

Potential Customers

Recommended actions:

- Premium product promotion
- Membership upgrades
- Personalized offers
- Targeted loyalty campaigns

Low-Engagement Customers

Recommended actions:

- Win-back campaigns
- Discount offers
- Personalized emails
- Limited-time promotions
- Feedback surveys

35. CUSTOMER RETENTION STRATEGY

Customer retention should focus particularly on customers showing declining engagement.

Important indicators include:

- Recency
- Purchase frequency
- Spending score
- Satisfaction score

A business can create automated customer campaigns based on these indicators.

For example:

High-value + recent purchase

→ Loyalty reward

High-value + long recency

→ VIP reactivation campaign

Low-value + high recency

→ Discount or product recommendation

Low satisfaction

→ Feedback and service recovery campaign

36. MODEL EVALUATION

The clustering model was evaluated using:

Elbow Method

Measures within-cluster variation using inertia.

Silhouette Score

Measures how well observations fit within their assigned clusters relative to neighboring clusters.

PCA Visualization

Provides a visual representation of cluster separation.

The exact silhouette score should be copied from:

```
silhouette_score(  
    X_scaled,  
    df["Cluster"]  
)
```

37. PROJECT OUTPUT

The final project produces a segmented customer dataset.

```
df.to_csv(  
    "customer_segmentation_with_clusters.csv",  
    index=False  
)
```

The final CSV contains:

- Customer information
- Behavioral information
- Cluster number
- Customer segment name
- PCA coordinates

This dataset can be used for further analysis or visualization in Power BI or other analytics tools.

38. LIMITATIONS

There are several limitations to the project.

1. Synthetic Dataset

The dataset used in this project is synthetic and was created for educational/project purposes.

Real customer data may produce different segmentation patterns.

2. K-Means Assumptions

K-Means is distance-based and works best when clusters are reasonably compact.

3. Number of Clusters

The final number of clusters depends on the selected evaluation criteria and business interpretability.

4. Feature Selection

Different features may produce different customer segments.

5. No Time-Series Modeling

The project uses customer-level behavioral variables rather than modeling customer behavior over time.

39. FUTURE SCOPE

The project can be improved in several ways.

Real Customer Data

The model can be trained using actual transaction data.

Advanced Clustering

Other algorithms could be compared:

- Hierarchical Clustering
- DBSCAN
- Gaussian Mixture Models

RFM Analysis

Recency, Frequency, and Monetary analysis could be integrated with machine learning.

Customer Lifetime Value

Customer lifetime value could be incorporated to identify the most profitable customer groups.

Predictive Analytics

A future model could predict:

- Customer churn
- Next purchase
- Customer lifetime value
- Product preference

Power BI Dashboard

The final segmentation can be integrated into an interactive dashboard containing:

- Customer segment distribution
- Revenue by segment
- Spending by segment
- Category preferences
- Membership distribution
- Customer location
- Segment-level KPIs

40. CONCLUSION

The Customer Segmentation project successfully demonstrates the application of unsupervised machine learning to customer analytics.

A dataset containing 1,000 customer records was analyzed using Python. The project included data cleaning, exploratory data analysis, feature selection, standardization, K-Means clustering, Elbow Method analysis, Silhouette Score evaluation, and PCA visualization.

Four customer segments were created using K-Means clustering. Each segment can be interpreted according to customer income, spending behavior, purchasing frequency, recency, satisfaction, and other behavioral characteristics.

The segmentation provides a practical foundation for targeted marketing and customer relationship management.

Businesses can use the results to:

- Identify high-value customers
- Retain important customers
- Reactivate low-engagement customers
- Personalize product recommendations
- Improve loyalty programs
- Increase customer lifetime value
- Develop targeted marketing campaigns

Overall, the project demonstrates how machine learning can transform raw customer data into actionable business insights.

41. FINAL DELIVERABLES

The completed project should contain the following files:

1. Dataset

customer_segmentation_1000.csv

2. Jupyter Notebook

customer_segmentation_project.ipynb

3. Final Segmented Dataset

customer_segmentation_with_clusters.csv

4. Final Project Report

Customer_Segmentation_Final_Report.pdf

43. FINAL PRESENTATION SUMMARY

For a project presentation, the project can be summarized as:

Problem:

Businesses have different types of customers and need targeted marketing strategies.

Solution:

Use K-Means clustering to automatically segment customers.

Dataset:

1,000 customer records with demographic and behavioral features.

Preprocessing:

Missing-value treatment, duplicate removal, feature selection, and standardization.

Model:

K-Means clustering.

Cluster Selection:

Elbow Method and Silhouette Score.

Visualization:

PCA and multiple customer behavior graphs.

Result:

Four customer segments suitable for targeted business strategies.

Business Value:

Improved personalization, customer retention, marketing efficiency, and customer lifetime value.

44. FINAL STATEMENT

This project demonstrates the complete machine-learning lifecycle from raw customer data to actionable customer segmentation. The resulting segmentation framework can help businesses understand customer behavior and make data-driven decisions for marketing, retention, personalization, and revenue growth.